Unit 1: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01 • Read conditions and nonlinearity

A device remains in the same state during two same-polarity readings: 2 µA at 0.1 V and 6 µA at 0.2 V. Calculate apparent conductances and test a linear, origin-passing resistor model. What conditions should accompany a resistance measurement?

E02 • Operations and polarity

Define SET, RESET, forming and read. In a specified bipolar device, a positive top-electrode voltage relative to the bottom electrode performs SET. Which polarity is used for RESET under this operating convention? Can it be generalized to other electrode stacks? Why limit programming current?

E03 • Evidence and mechanisms

A metal/oxide/metal device shows reproducible bipolar hysteresis. Does this alone establish ECM or VCM? Identify possible mobile species and two types of distinguishing evidence, with limitations.

E04 • Ideal crossbar

Let row voltages be $V = (0.2, 0.1)$ V and conductances $G_{11} = 2$, $G_{12} = 1$, $G_{21} = 1$, $G_{22} = 3$ µS. Assume linear devices, zero wire resistance and virtual-ground columns. Find both column currents and every branch current. If wire drop instead leaves 0.08 V at every crosspoint in row 2, with other conditions unchanged, recalculate and explain whether the original ideal formula still applies.

E05 • Read margin

At a specified operating point, HRS currents range from 1 to 3 µA and LRS currents from 5 to 9 µA. Define margin as minimum LRS current minus maximum HRS current. Calculate it and give thresholds that distinguish the closed intervals using the rule “strictly greater than threshold means LRS.” If newly observed distributions overlap, is an error rate determined by this information alone?

E06 • Reliability claims

Two tests report 10^6 cycles and ten years of retention. List at least four conditions required for comparison, distinguish cycle-to-cycle and device-to-device variability, and explain why one hour at 85 °C alone does not establish ten years at room temperature.

Unit 1: Exercises (continued)

E07 • PCM heating and cooling

In independent trials assume a lumped thermal model with ambient 300 K, $R_{th} = 10^5$ K/W and $\tau = 10$ ns. Starting at 300 K, calculate the temperature after a 4 mW pulse lasting 30 ns and after a 10 mW pulse lasting 10 ns. In the second trial, power then becomes zero and parameters remain constant; find the time to cool to 600 K. If illustrative crystallization and melting thresholds are 600 K and 900 K, what additional information is needed before concluding either pulse produced a stable phase?

E08 • Read-and-write OR

Two parallel cells encode 0 as $1\,\mu$S and 1 as $10\,\mu$S. Assume ideal 0.1 V non-disturbing linear reads, a virtual-ground current sensor and ideal comparator: output 1 iff the total current is strictly greater than $0.6\,\mu$A. Compute all four currents and the truth table. Describe how a separately initialized output cell could store the result, and identify the digital circuit needed.

E09 • Dynamic synapse

Suppose $dg/dt = -g/\tau$ between instantaneous spikes; each spike adds $2\,\mu$S. Let $g(0^-) = 0$, $\tau = 10$ ms, and spikes arrive at 0 and 10 ms. Compute $g(10^-)$, $g(10^+)$, $g(20)$ and the current immediately after the second spike under an ideal nondisturbing 0.1 V read. Is the exponential law necessarily valid for all devices? Can this short-term response establish long-term nonvolatile retention?

E10 • Biased random bits

Assume raw bits are independent and identically distributed with $P(1) = 0.6$. Calculate single-bit min-entropy and the output probabilities and average yield per raw bit of nonoverlapping von Neumann pairs, using 01→0, 10→1 and discarding 00 and 11. Why are bias alone and finite statistical tests insufficient to establish entropy rate or cryptographic unpredictability?

E11 • An implication gate with a stated switching window

Two memory cells P, Q share a lower node connected to ground through $G_L = 3\,\mu$S. Their upper terminals are held at 0.5 V and 1 V, respectively, for a specified pulse. Each initial state is modelled as a linear conductance: HRS (logic 0) is $1\,\mu$S and LRS (logic 1) is $10\,\mu$S. An HRS Q SETs to LRS if its positive voltage is at least 0.65 V; otherwise it stays in its initial state. Assume P is undisturbed for $|V_P^{dev}| < 0.3$ V and Q remains latched after switching. For all four input states compute the initial common-node voltage, the voltages across P and Q, the final Q state and the truth function. Recompute the node voltage after a 00 input switches, and state why the model is not a universal device guarantee.

Unit 1: Exercises (continued)

E12 • Multiplication with bit streams

For independent bit pairs let $P(A_i = 1) = 0.6$, $P(B_i = 1) = 0.5$ and $Y_i = A_i \wedge B_i$. Compute $P(Y_i = 1)$, the expected number of ones in 100 independent identically distributed output bits and its standard deviation. Give a counterexample showing that AND need not multiply the marginal probabilities when the two streams are correlated, and describe the effect of temporal dependence on the count variance.

Lin Cing
https://github.com/xinhi6716/resistive-memory-self-study